

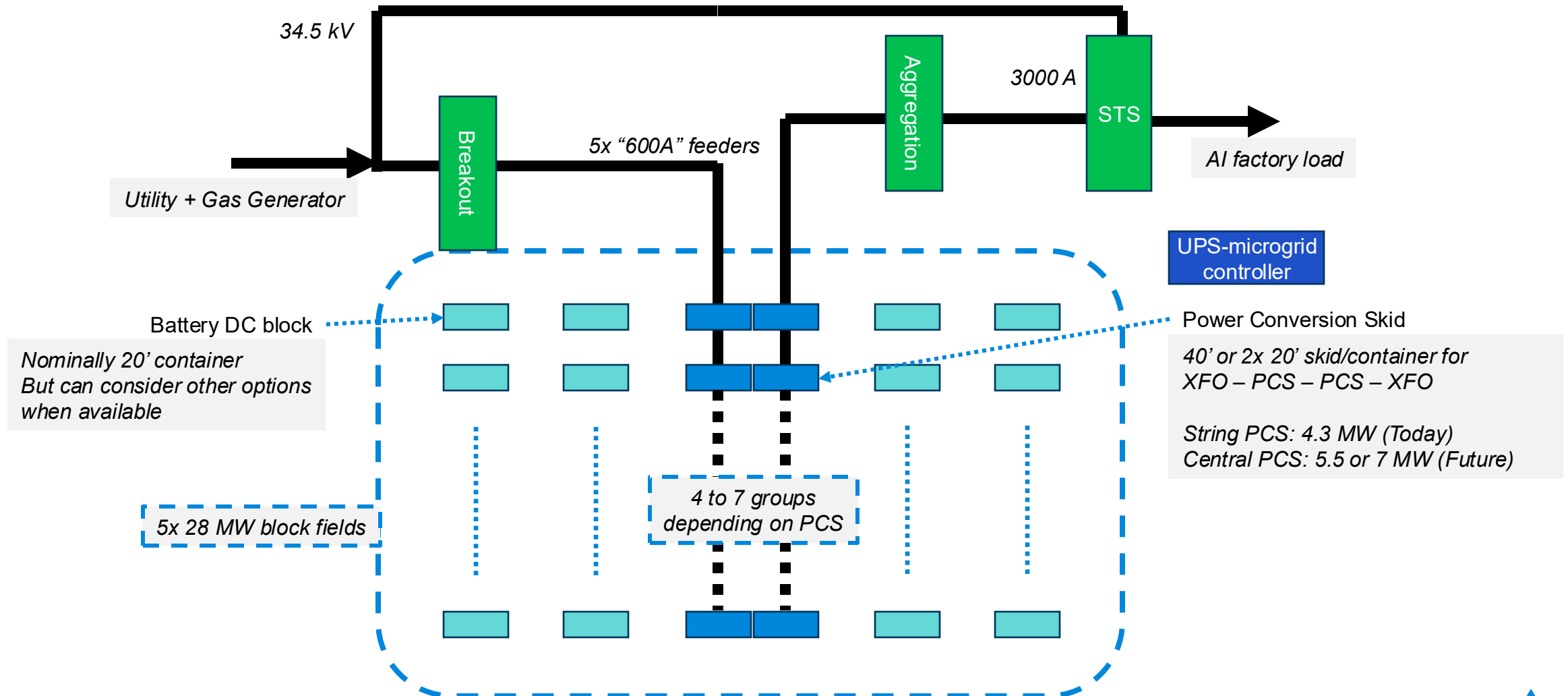
2026-04-30

140 MW “UPS” with 4-hr storage

Solution options



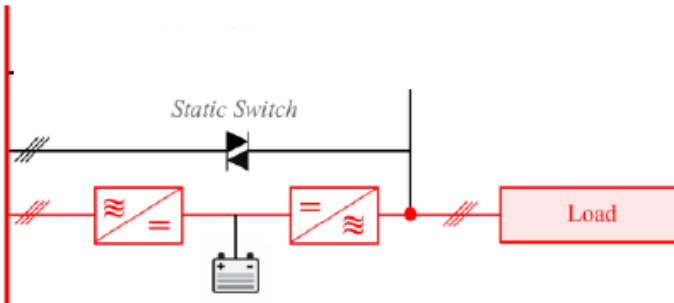
Double-conversion UPS: 1 possible implementation



Modes of operation and requirements

Double Conversion Mode

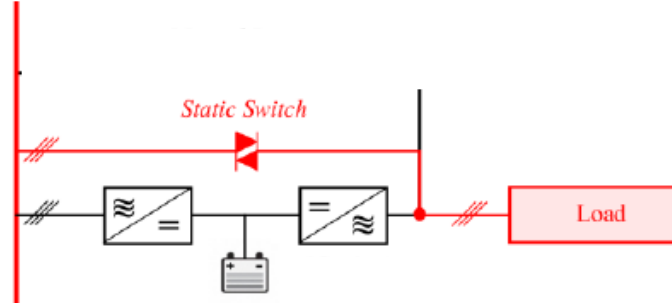
During AI training sessions



- Output-side converter shall regulate load voltage and frequency within 5% of nominal under load step condition of 20% to 90% of rated power within 16 ms. (1 ac cycle)
- Input-side converter shall draw smooth active power from the grid (variation over minutes rather than seconds – see next page) and be capable of reactive power support if required.

Bypass Mode

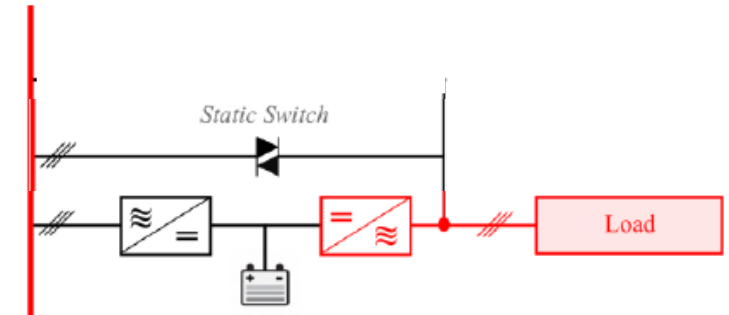
During AI inference sessions



- Input converter, output converter, or both converters operate in STATCOM mode.
- Voltage regulation and power factor control are required.

Backup mode

During utility / onsite generator outage

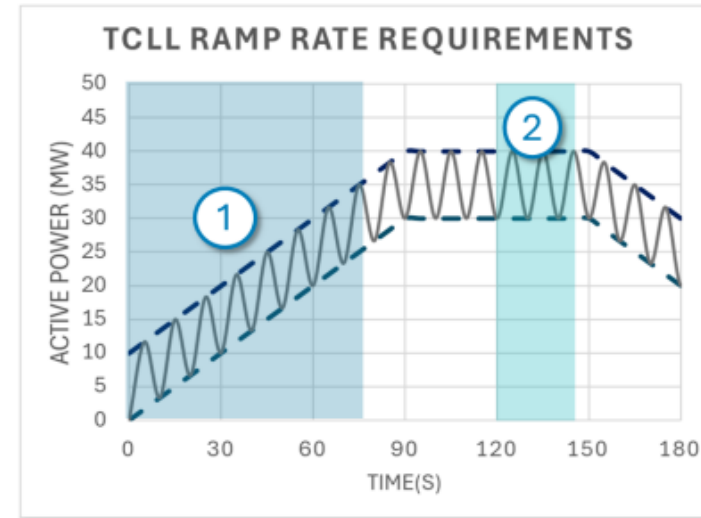


- Input-side converter disconnected from grid.
- Output-side converter shall regulate load voltage and frequency within 5% of nominal under load step condition of 20% to 90% of rated power within 16 ms.

- Target transition time between modes within a quarter of a cycle (~4ms)
- Target valid for planned transitions between Double-conversion and Bypass modes as well as for unplanned transition from Bypass mode to Backup mode.
- Transitions may be initiated by SCADA system

Input side requirements

- Power drawn from utility must meet ramp rate and intra-minute requirements
- Should be easier with double-conversion UPS
- Can single-conversion UPS achieve this?
- Additional references:
 - [BESS Self-Qualification Guidelines | NVIDIA](#)
 - [SOCO Technical Requirements for TCLLs - v1.2 2025-12-01 \(1\).pdf](#)



Technical Requirements for Transmission Connected Large Loads

Effective December 1st, 2025

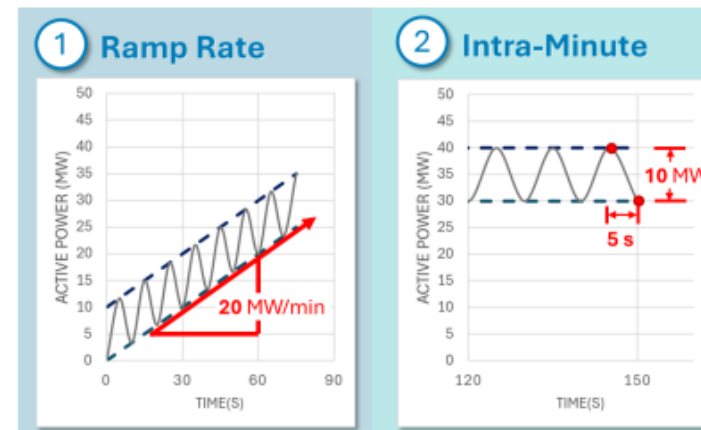


Figure 3. Visual of Ramp Rate and Intra-Minute Criteria